Abubakar Khawaja

Senior Frontend Engineer | React | Next.js | TypeScript | React Native

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PROFESSIONAL SUMMARY

Frontend Engineer with 5+ years of experience architecting and shipping responsive, high-performance web applications using React, Next.js and TypeScript, and cross-platform mobile apps with React Native and Expo. I prioritize fast renders, scalable component systems, and intuitive interfaces that hold up under product growth, and I have built AI-driven frontends that stream model output and surface agentic retrieval to the user. I am comfortable owning UI architecture and design systems, and I thrive on partnering with cross-functional teams to deliver exceptional user experiences.

EXPERIENCE

Senior Software Engineer | Codeupnow Sept 2025 – Present

- Built reusable component libraries with Storybook, keeping the UI consistent across several projects.
- Added WebSockets for real-time updates in live data processing.
- Optimized frontend performance with lazy loading, code splitting, and image optimization.
- Mentored junior engineers through regular code reviews and drove Agile feature delivery in tight collaboration with UX/UI designers and backend teams.
- Worked with other teams to pin down requirements and ship features on schedule.

Senior Frontend Engineer | Arbisoft July 2021 – Apr 2026

- Led cross-functional framework migrations (Material UI v4 to v5, React Router v4 to v5) and architected reusable Storybook component libraries to ensure design consistency.
- Redesigned React frontend architecture, reducing unnecessary re-renders by 30% and improving page load times by 20-30% via memoization, code-splitting, and optimized state management.
- Implemented SSR, SSG, and ISR using Next.js while improving Core Web Vitals, accessibility, SEO, and overall application responsiveness.
- Elevated automated test coverage from 10% to over 60% (Jest, Cypress, Vitest) and integrated quality gates (ESLint, Husky), reducing production bugs by 25%.
- Reworked GitLab CI/CD pipelines with parallel jobs and layer caching, reducing deployment times by 40%.
- Used AI coding tools such as Cursor, Claude Code, Codex, GitHub Copilot, and ChatGPT for both autonomous multi-step tasks and hands-on pair-programming, applying them to PR review, architecture discussions, and custom internal tooling, while comparing model outputs using structured prompts.

SELECTED PROJECTS

Landit, Career Platform · *React + Next.js + TypeScript*

- Built pages with React Server Components on the Next.js App Router, keeping data fetching on the server and reserving client components for the interactive parts of the page.
- Worked across both the Next.js App Router and Pages Router during the project's router migration.
- Migrated Formik to React Hook Form, improving performance and maintainability.

Meet Gabi, Social Discovery Platform · *React + Django · Codeupnow*

- Built and maintained full-stack features for a social discovery platform, with React on the frontend and Django on the backend.
- Integrated real-time messaging and user analytics.

FixAlert, Maintenance Platform · *React + TypeScript*

- Revamped the UI of a computerized maintenance-management system in an Agile environment.
- Built reporting metric views and optimized frontend data fetching with React Query.
- Untangled a filter set where changing any one filter wiped the rest, so a change now only touches the filters that genuinely depend on it, and selections stay put.

Retrieval Chat, Agentic AI Assistant · *React + Vercel AI SDK · Personal Project*

- Built a streaming chat interface with the Vercel AI SDK, rendering model output token-by-token with markdown formatting and sanitised HTML.
- Ran object detection client-side with Transformers.js, drawing confidence-filtered bounding boxes in the browser.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3/SCSS
Frontend: React.js, Next.js (App Router, Pages Router), Vue.js, TypeScript, Redux Toolkit, TanStack Query, React Hook Form, Tailwind CSS, Material UI, SCSS
Mobile: React Native, Expo, NativeWind, Expo Router, Flutter, Dart
State & Data: Redux Toolkit, Zustand, Context API, TanStack Query, WebSockets, RESTful APIs
Backend Integration: REST APIs, GraphQL, API Design Collaboration

Frontend Engineering: Performance Profiling, Code Splitting, Lazy Loading, Accessibility (WCAG), Responsive Design, Component Architecture
Testing & Quality: Jest, Cypress, Vitest, Storybook, ESLint, Prettier, Husky (pre-commit hooks)
Build & Tooling: Vite, Git, GitLab CI/CD, GitHub Actions, Docker
Integrations: Stripe (payments), SendGrid & Resend (email), Firebase Auth, Google Maps, PostHog & Google Analytics, Google Looker (BI), Grafana, JWT/OAuth2
AI/ML: LLM Application Architecture, AI Agents, Tool/Function Calling, Structured Outputs (Zod, Pydantic), Model Context Protocol (MCP), RAG, Embeddings & Vector Databases (pgvector), Prompt & Context Engineering, OpenAI / OpenRouter / Ollama (local models), Vercel AI SDK

EDUCATION

B.S. Software Engineering Punjab University College of Information & Technology (PUCIT) · GPA: 3.08

CERTIFICATIONS

- The AI Engineer Path & Prompt Engineering for Web Developers — Scrimba (July 2026)